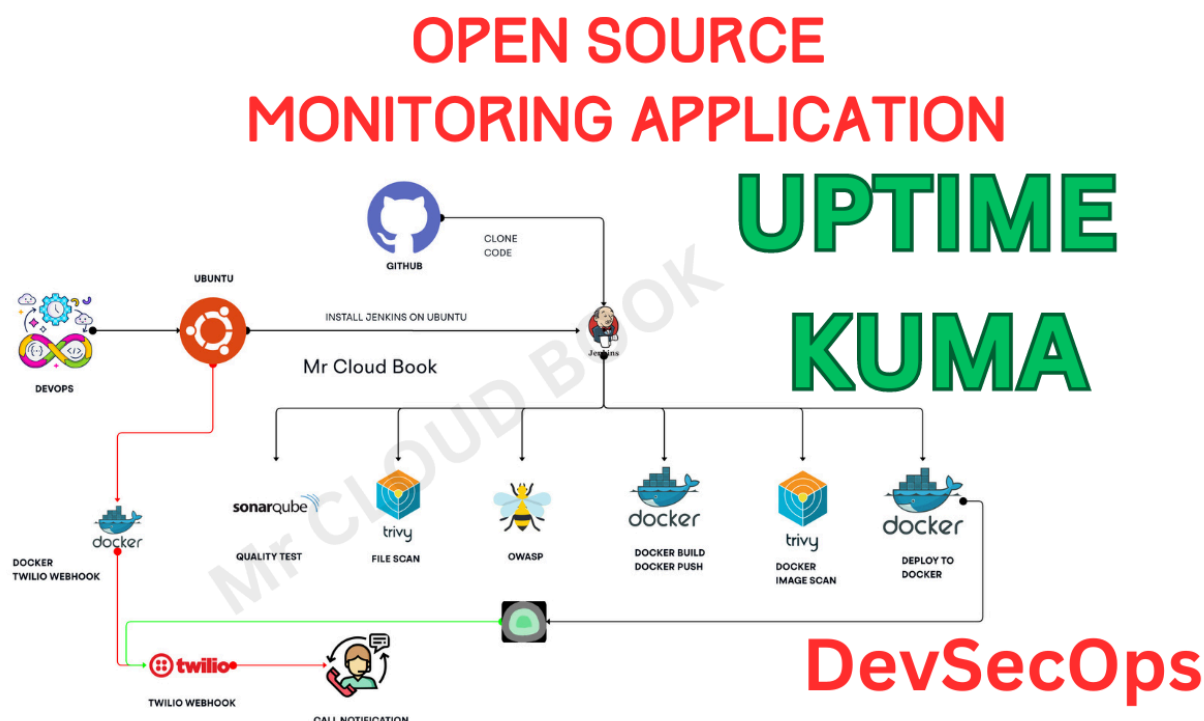


DevOps

CI/CD Pipeline for Uptime-Kuma | Get Real-Time Call Alerts for Server Downtime | Open Source Monitoring



mrcloudbook.com · 26 June 2024



Uptime-Kuma is an open-source self-hosted monitoring tool that provides a sleek interface to track the uptime of websites, services, and APIs. When integrated into a CI/CD pipeline for a blog, it helps ensure continuous availability and performance

monitoring. Here's a detailed explanation of how Uptime-Kuma can be utilized effectively in this context:

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What is Uptime-Kuma?

- **Overview:** Uptime-Kuma is a self-hosted monitoring tool designed to track the uptime and performance of websites, services, and APIs.
- **Features:**
 - Real-time monitoring with customizable intervals.
 - Alerts via multiple channels (email, Slack, Telegram, etc.).
 - Historical data and uptime statistics.
 - Sleek, user-friendly interface.

Now, let's get started and dig deeper into each of these steps:-

1. AWS Account and Login

- If you don't have an AWS account, sign up at [AWS Sign Up](#).
- Log in to the AWS Management Console.

2. Launch an EC2 Instance

A. Navigate to EC2 Dashboard:

- Go to the [EC2 Dashboard](#).

B. Choose an Amazon Machine Image (AMI):

- Click “Launch Instance.”
- In the AMI selection screen, search for “Ubuntu 24.04”.

C. Choose an Instance Type:

- Select the `t2.large` instance type.
- Click “Next: Configure Instance Details.”

D. Configure Instance Details:

- Use the default settings.
- Ensure you have selected the correct VPC and subnet.
- Click “Next: Add Storage.”

E. Add Storage:

- Modify the root volume size to 30 GB.
- Click “Next: Add Tags.”

F. Add Tags:

- (Optional) Add tags for better management.
- Click “Next: Configure Security Group.”

G. Configure Security Group:

- Create a new security group.
- Set the security group name and description.

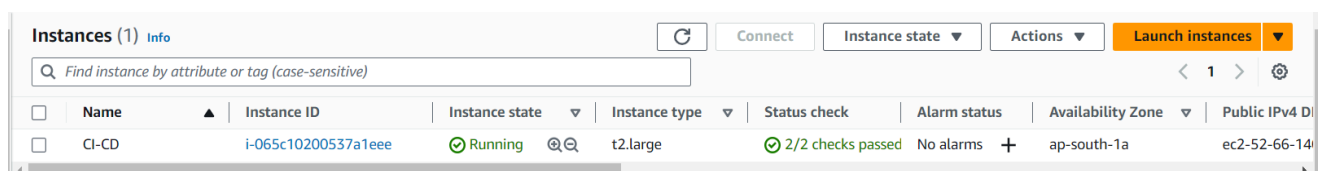
- Add a rule to allow all traffic (not recommended for production environments):
 - Type: All Traffic
 - Protocol: All
 - Port Range: All
 - Port Range : 22, 80, 443, 8080, 9000, 3001, 5000
 - Source: 0.0.0.0/0 (for IPv4) and ::/0 (for IPv6)
- **Warning:** Opening all ports is insecure and should only be done for learning purposes in a controlled environment.
- Click “Review and Launch.”

H. Review Instance Launch:

- Review all configurations.
- Click “Launch.”

I. Select/Create a Key Pair:

- Select “Create a new key pair.”
- Name the key pair.
- Download the key pair file (.pem) and store it securely.
- Click “Launch Instances.”



Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 D
CI-CD	i-065c10200537a1eee	Running	t2.large	2/2 checks passed	No alarms	ap-south-1a	ec2-52-66-14

Install Jenkins, Docker and Trivy

To Install Jenkins

Connect to your console, and enter these commands to Install Jenkins



```
vi jenkins.sh
```



```
#!/bin/bash
sudo apt update -y
wget -O - https://packages.adoptium.net/artifactory/api/gpg/key/public
echo "deb [signed-by=/etc/apt/keyrings/adoptium.asc] https://packages.adoptium.net/artifactory/debian [arch=amd64] *" | sudo tee /etc/apt/sources.list.d/adoptium.list > /dev/null
sudo apt update -y
sudo apt install temurin-17-jdk -y
/usr/bin/java --version
curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/keyrings/jenkins-keyring.asc > /dev/null
sudo apt-get update -y
sudo apt-get install jenkins -y
sudo systemctl start jenkins
sudo systemctl status jenkins
```



```
sudo chmod 777 jenkins.sh
sudo su    #move into root and run
./jenkins.sh    # this will install jenkins
```



Once Jenkins is installed, you will need to go to your AWS EC2 Security Group and open Inbound Port 8080, since Jenkins works on Port 8080.

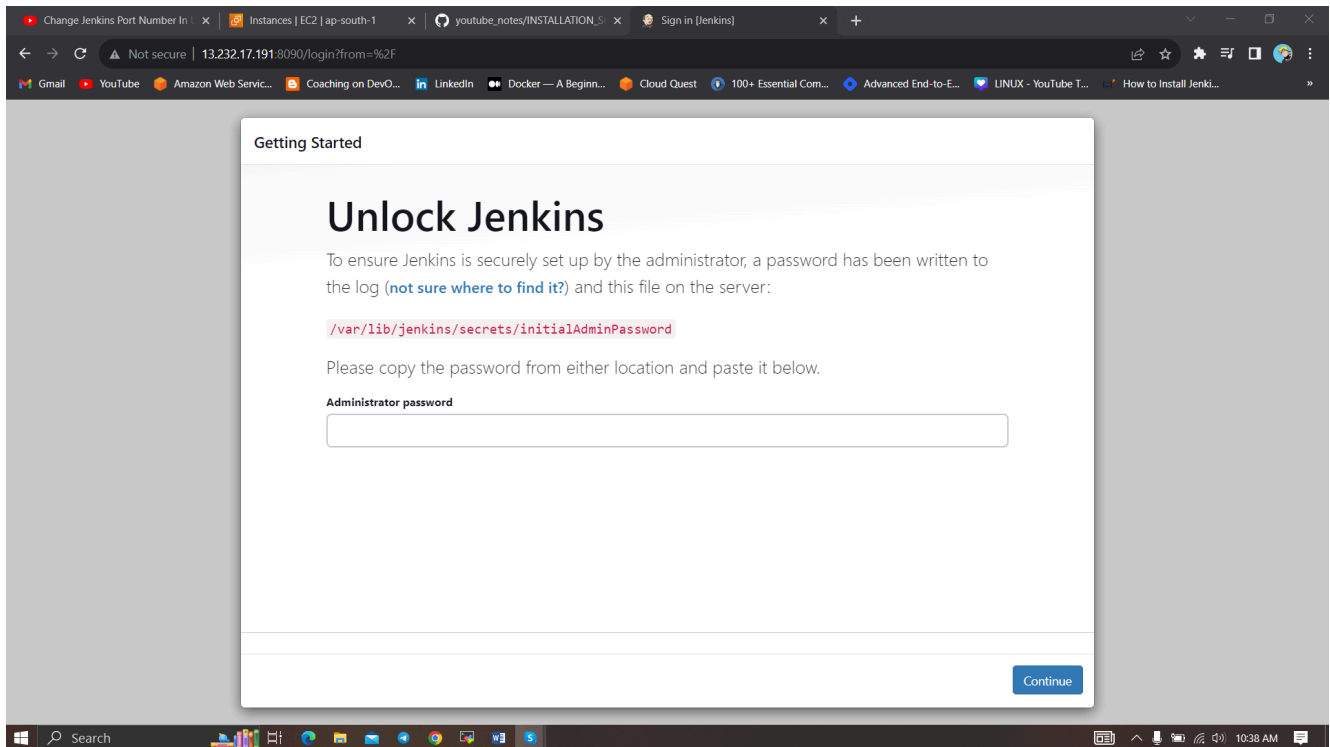
Now, grab your Public IP Address



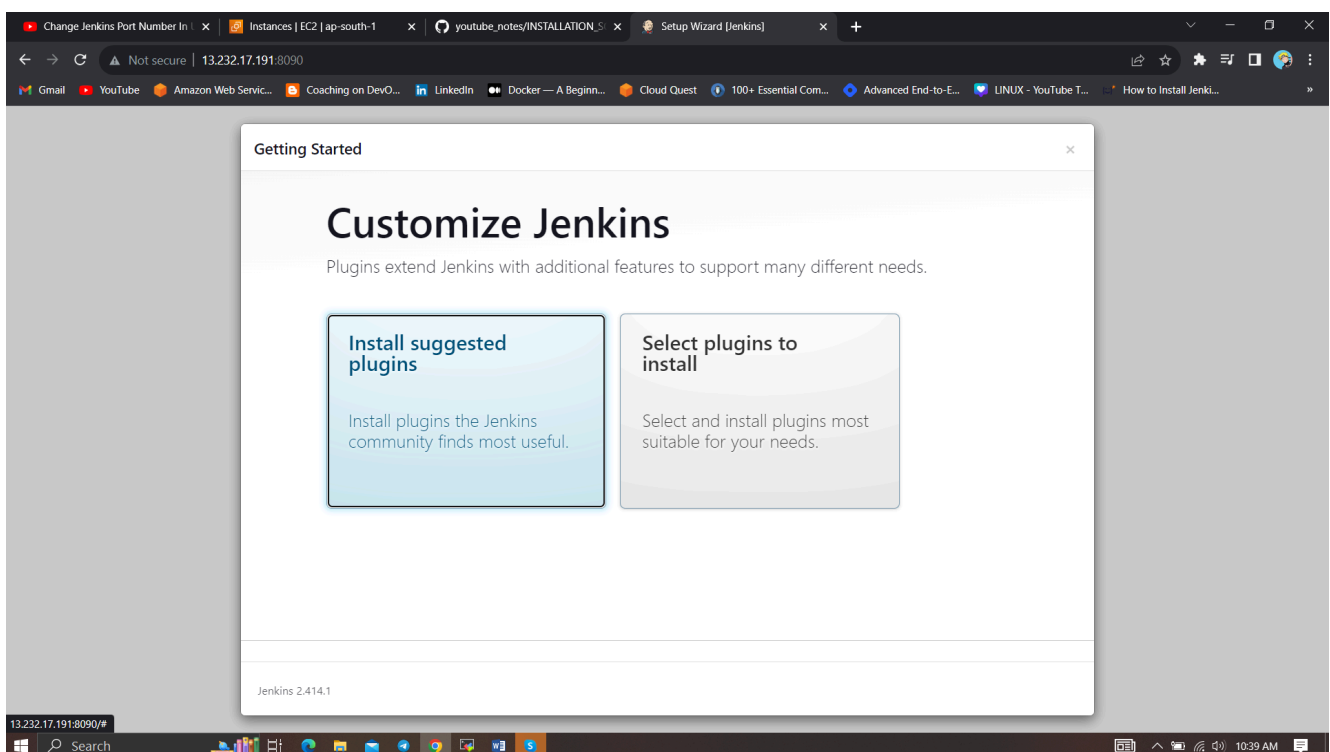
EC2 Public IP Address:8080

In Jenkins server

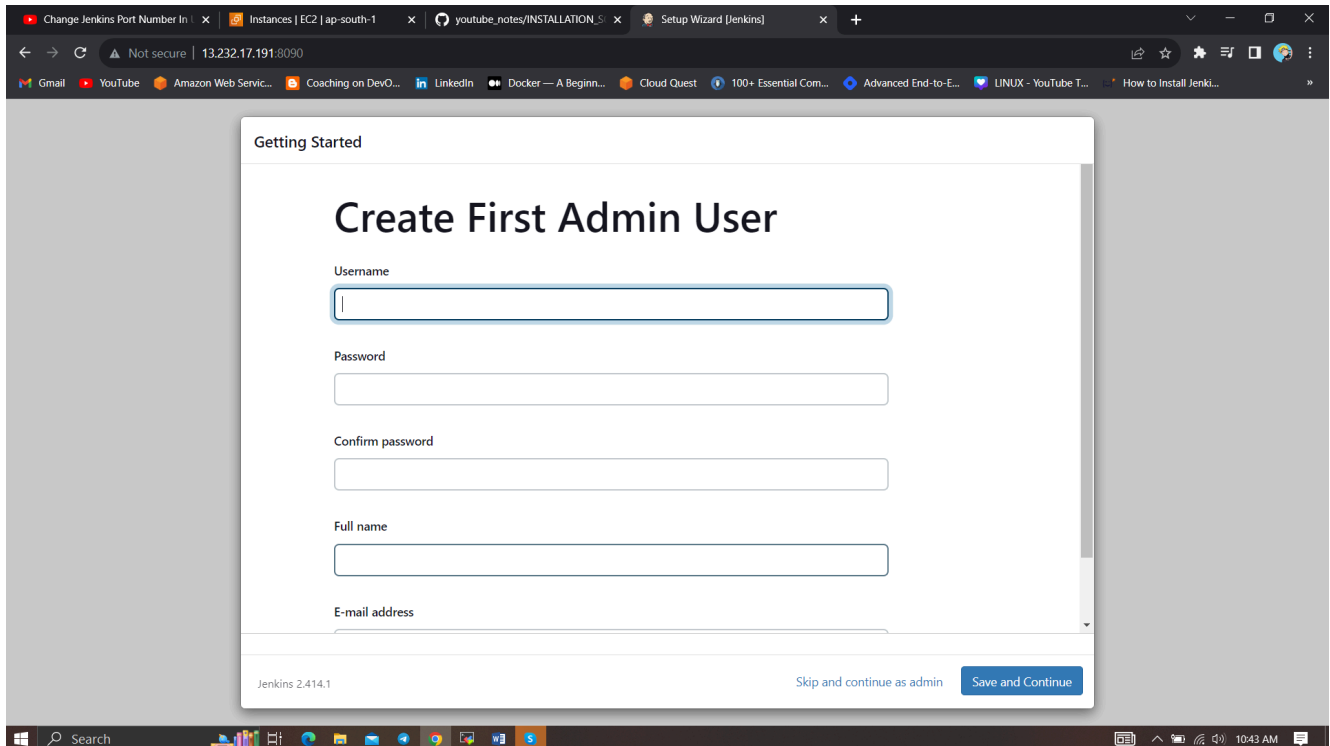
```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```



Unlock Jenkins using an administrative password and install the suggested plugins.

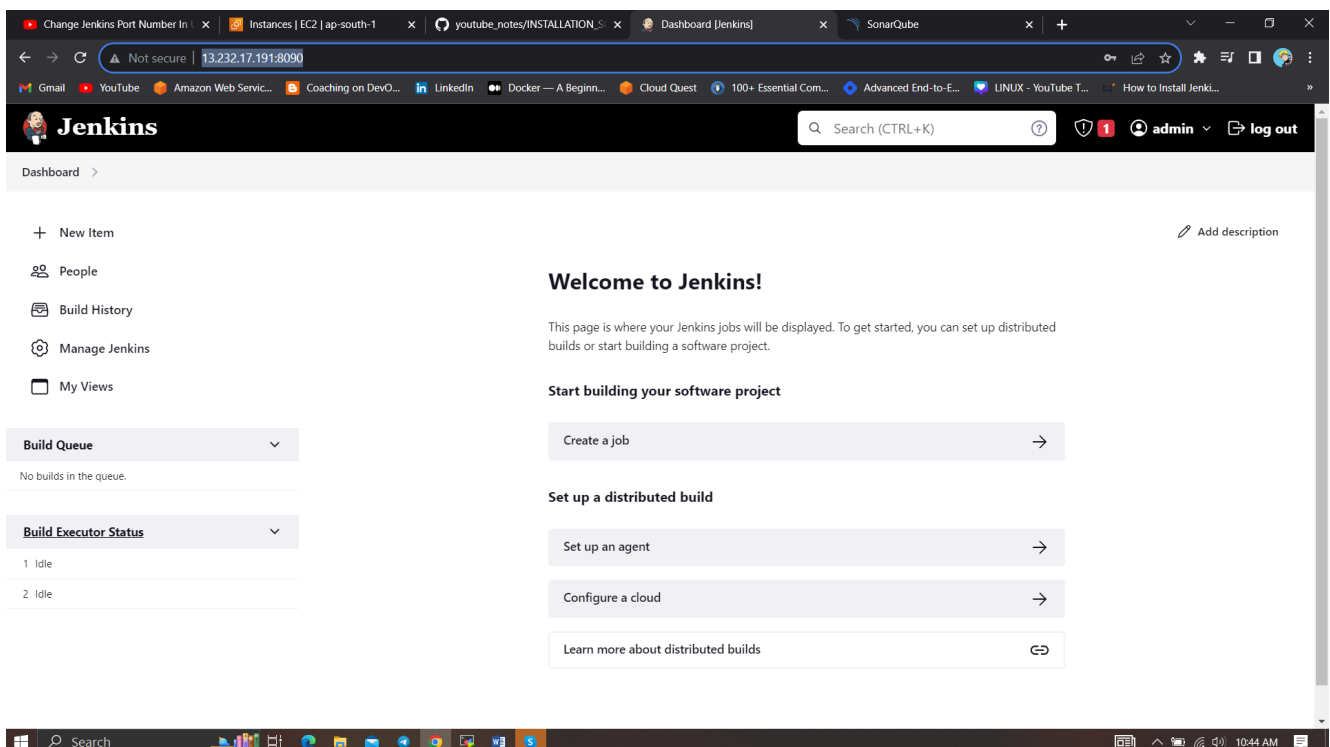


Jenkins will now get installed and install all the libraries.



Create a user click on save and continue.

Jenkins Getting Started Screen.



Install Docker



```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt
sudo chmod a+r /etc/apt/keyrings/docker.asc
# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/d
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list >/dev/null
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx
sudo usermod -aG docker ubuntu
newgrp docker
sudo chmod 777 /var/run/docker.sock
```



After the docker installation, we create a sonarqube container (Remember to add 9000 ports in the security group).

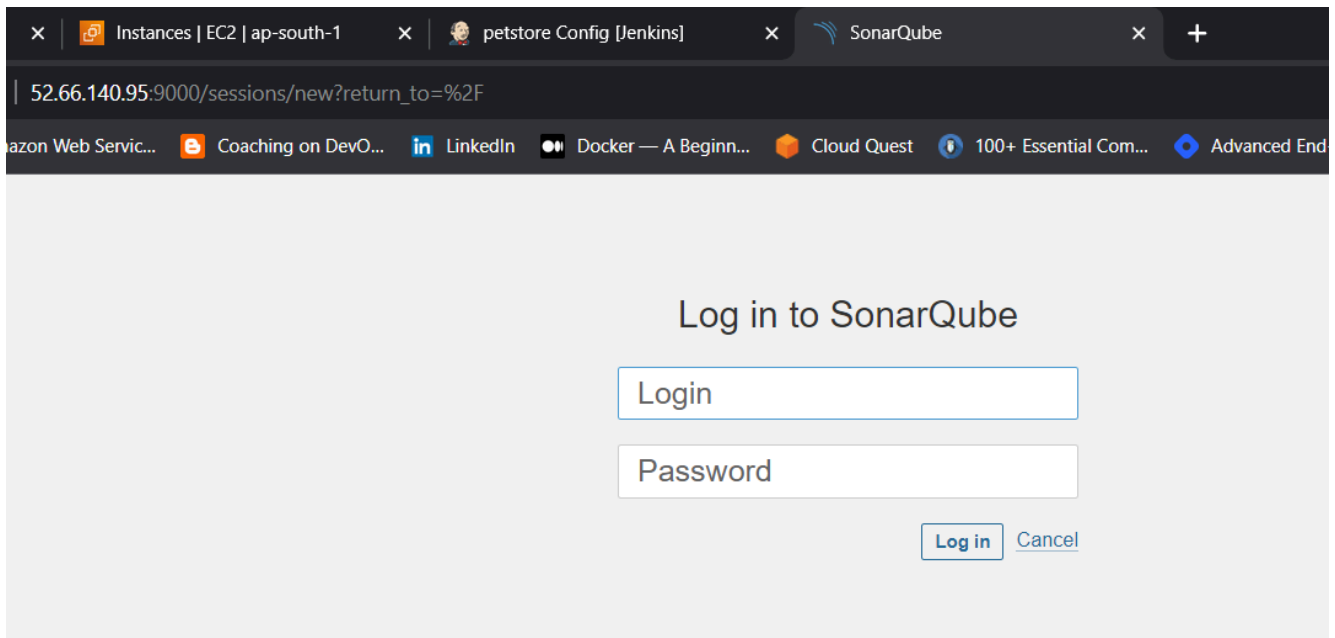


```
sudo chmod 777 /var/run/docker.sock
docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
```



```
ubuntu@ip-172-31-42-253:~$ sudo chmod 777 /var/run/docker.sock
ubuntu@ip-172-31-42-253:~$ docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
Unable to find image 'sonarqube:lts-community' locally
lts-community: Pulling from library/sonarqube
44ba2882f8eb: Pull complete
2cabec57fa36: Pull complete
c29481384b6a: Pull complete
bf7b17ee74f8: Pull complete
38617faac714: Pull complete
706f20f58f5e: Pull complete
65a29568c257: Pull complete
Digest: sha256:1a118f8ab960d6c3d4ea8b4455a5a6560654511c88a6816f1603f764d5dcc77c
Status: Downloaded newer image for sonarqube:lts-community
4b60c96bf9ad3d62289436af7f752fdb04993092d0ca3065e2f2e32301b50139
ubuntu@ip-172-31-42-253:~$ docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
4b60c96bf9ad   sonarqube:lts-community  "/opt/sonarqube/dock..."  9 seconds ago  Up 5 seconds  0.0.0.0:9000->9000/tcp, :::9000->9000/tcp  sonar
ubuntu@ip-172-31-42-253:~$
```

Now our Sonarqube is up and running

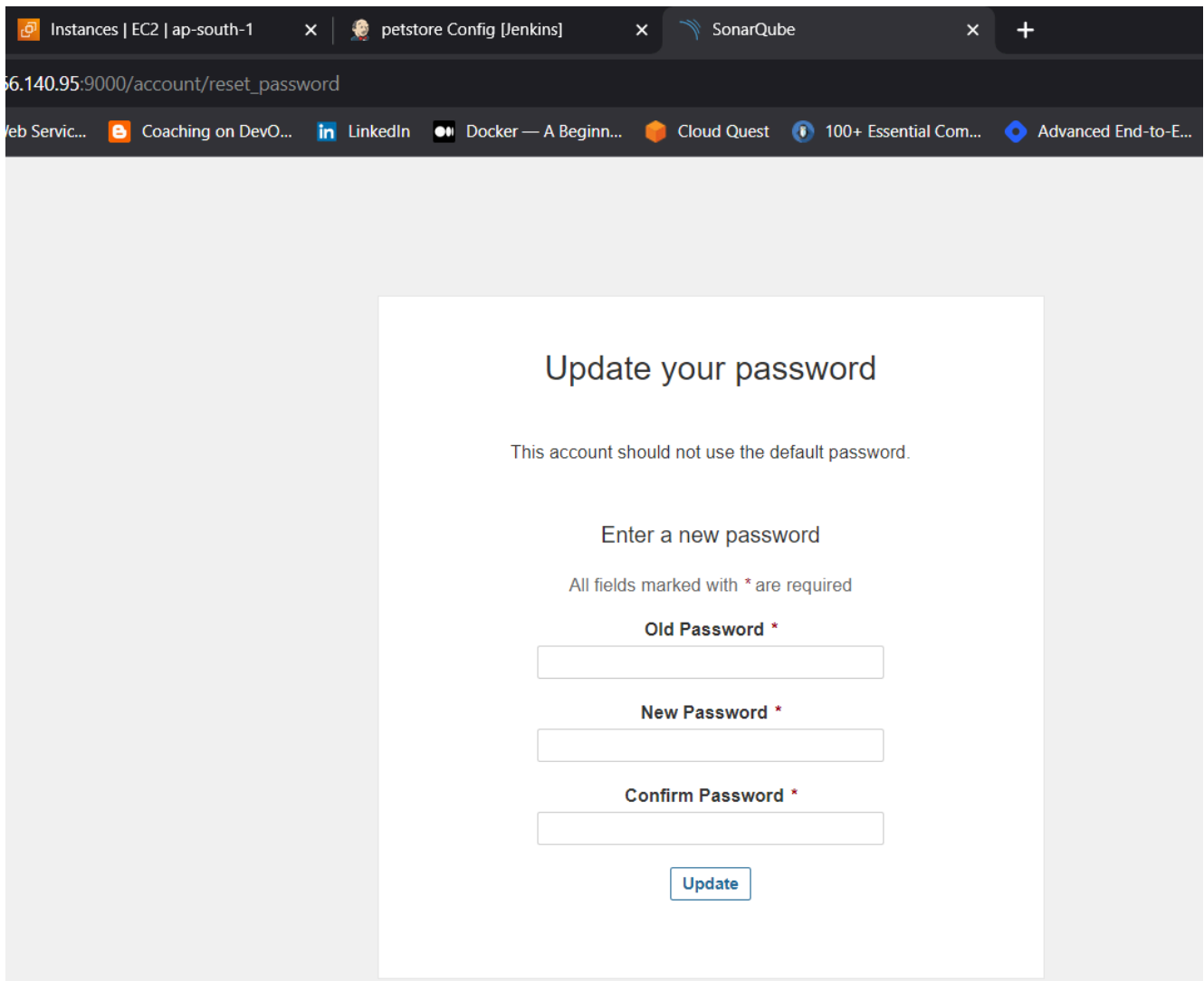


Enter username and password, click on login and change password



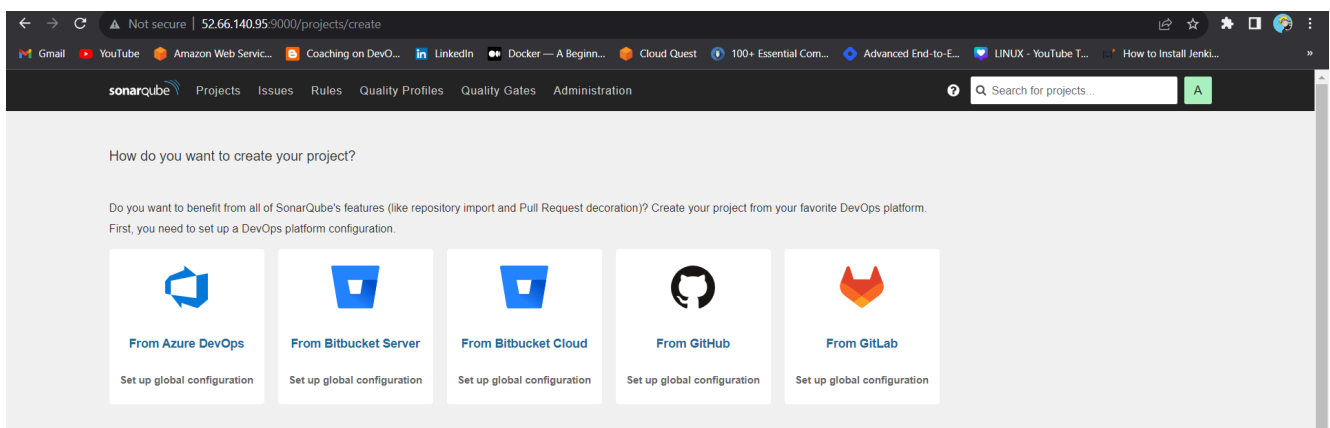
```
username admin  
password admin
```





The screenshot shows a web browser window with the URL `52.66.140.95:9000/account/reset_password`. The browser tabs include 'Instances | EC2 | ap-south-1', 'petstore Config [Jenkins]', and 'SonarQube'. The page title is 'Update your password'. Below the title, it says 'This account should not use the default password.' and 'Enter a new password'. A note states 'All fields marked with * are required'. There are three input fields: 'Old Password *', 'New Password *', and 'Confirm Password *'. An 'Update' button is at the bottom.

Update New password, This is Sonar Dashboard.



The screenshot shows the SonarQube 'Create project' page. The URL is `52.66.140.95:9000/projects/create`. The page has a navigation bar with 'sonarqube', 'Projects', 'Issues', 'Rules', 'Quality Profiles', 'Quality Gates', and 'Administration'. A search bar is present. The main content area asks 'How do you want to create your project?' and provides instructions: 'Do you want to benefit from all of SonarQube's features (like repository import and Pull Request decoration)? Create your project from your favorite DevOps platform. First, you need to set up a DevOps platform configuration.' Below this are five options: 'From Azure DevOps', 'From Bitbucket Server', 'From Bitbucket Cloud', 'From GitHub', and 'From GitLab'. Each option includes the text 'Set up global configuration'.

Install Trivy



```
vi script.sh
```

```
sudo apt-get install wget apt-transport-https gnupg lsb-release -y
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | g
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity
sudo apt-get update
sudo apt-get install trivy -y
```

Give permissions and run script

```
sudo chmod 777 script.sh
./script.sh
```

Next, we will log in to Jenkins and start to configure our Pipeline in Jenkins

Install Plugins like JDK, Sonarqube Scanner, NodeJs, OWASP Dependency Check

Install Plugin

Goto Manage Jenkins → Plugins → Available Plugins →

Install below plugins

Blue ocean

1 → Eclipse Temurin Installer

2 → SonarQube Scanner

3 → NodeJs Plugin

4 → Docker

5 → Docker commons

6 → Docker pipeline

7 → Docker API

8 → Docker Build step

9 → Owasp Dependency Check

Install	Name ↓	Released
☒	Eclipse Temurin installer 1.5 Provides an installer for the JDK tool that downloads the JDK from https://adoptium.net	1 yr 8 mo ago
☒	SonarQube Scanner 2.17.2 External Site/Tool Integrations Build Reports This plugin allows an easy integration of SonarQube , the open source platform for Continuous Inspection of code quality.	4 mo 7 days ago
☒	NodeJS 1.6.1 npm NodeJS Plugin executes NodeJS script as a build step.	10 mo ago
☒	OWASP Dependency-Check 5.5.0 Security DevOps Build Tools Build Reports This plug-in can independently execute a Dependency-Check analysis and visualize results. Dependency-Check is a utility that identifies project dependencies and checks if there are any known, publicly disclosed, vulnerabilities.	4 mo 1 day ago
☒	Docker 1.6.2 Cloud Providers Cluster Management docker This plugin integrates Jenkins with Docker	22 days ago
☒	Docker Commons 439.va_3cb_0a_6a_fb_29 Library plugins (for use by other plugins) docker Provides the common shared functionality for various Docker-related plugins.	11 mo ago
☒	Docker Pipeline 580.vc0c340686b_54 pipeline DevOps Deployment docker Build and use Docker containers from pipelines.	1 mo 5 days ago
☒	Docker API 3.3.6-90.ve7c6c7535ddd Library plugins (for use by other plugins) docker This plugin provides docker-java API for other plugins. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	23 days ago
☒	docker-build-step 2.12 Build Tools docker This plugin allows to add various docker commands to your job as build steps. Warning: This plugin version may not be safe to use. Please review the following security notices: • CSRF vulnerability and missing permission check	29 days ago

Configure Java and Nodejs in Global Tool Configuration

Goto Manage Jenkins → Tools → Install JDK(17) and NodeJs(18)→ Click on Apply and Save

JDK installations

Add JDK

≡ **JDK**

Name

jdk17

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☒ Install automatically ?

≡ **Install from adoptium.net** ?

Version ?

jdk-17.0.11+9 ▼

Add Installer ▼

Add NodeJS

NodeJS

Name

node18 **MR CLOUD BOOK**

☒ Install automatically ?

Install from nodejs.org

Version

NodeJS 18.0.0

For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail

☐ Force 32bit architecture

Global npm packages to install

Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax `packageName@version`

Global npm packages refresh hours

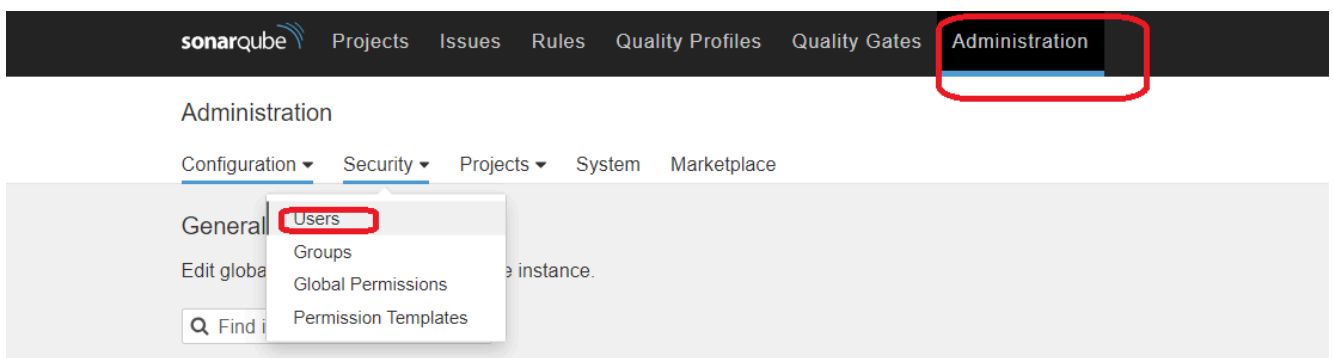
Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache

72

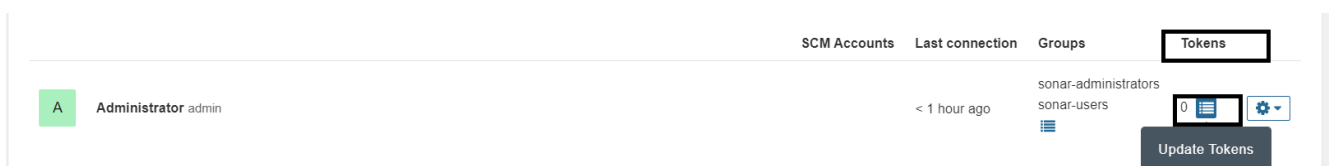
Add Installer ▾

Grab the Public IP Address of your EC2 Instance, Sonarqube works on Port 9000, so <Public IP>:9000. Goto your Sonarqube Server.

Click on Administration → Security → Users → Click on Tokens and Update Token → Give it a name → and click on Generate Token



click on update Token



Create a token with a name and generate

Tokens of Administrator

Generate Tokens

Name Expires in 30 days Generate

New token "Jenkins" has been created. Make sure you copy it now, you won't be able to see it again!

Copy `squ_21d162904c1c72cf8b39665f96480185c99dc2f9`

Name	Type	Project	Last use	Created	Expiration	
Jenkins	User		Never	September 8, 2023	October 8, 2023	Revoke

copy Token

Go to Jenkins Dashboard → Manage Jenkins → Credentials → Add Secret Text. It should look like this

Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >

New credentials

Kind Secret text

Scope Global (Jenkins, nodes, items, all child items, etc)

Secret **POST THE TOKEN HERE**

ID **Sonar-token**

Description **Sonar-token**

Create

You will this page once you click on create

Credentials that should be available irrespective of domain specification to requirements matching.

ID	Name	Kind	Description	
Sonar-token	sonar	Secret text	sonar	

Now, go to Dashboard → Manage Jenkins → System and Add like the below image.

Search for SonarQube Installation

SonarQube installations

List of SonarQube installations

Name

sonar-server

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Server URL

Default is <http://localhost:9000>

<http://65.0.93.191:9000>

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

Sonar-token

+ Add ▼

Click on Apply and Save

The **Configure System option** is used in Jenkins to configure different server

Global Tool Configuration is used to configure different tools that we install using Plugins

We will install a sonar scanner in the tools.

Manage Jenkins -> Tools -> SonarQube Scanner

☰ SonarQube Scanner

Name

sonar-scanner

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☒ Install automatically ?

☰ Install from Maven Central

Version

SonarQube Scanner 6.0.0.4432

Add Installer ▼

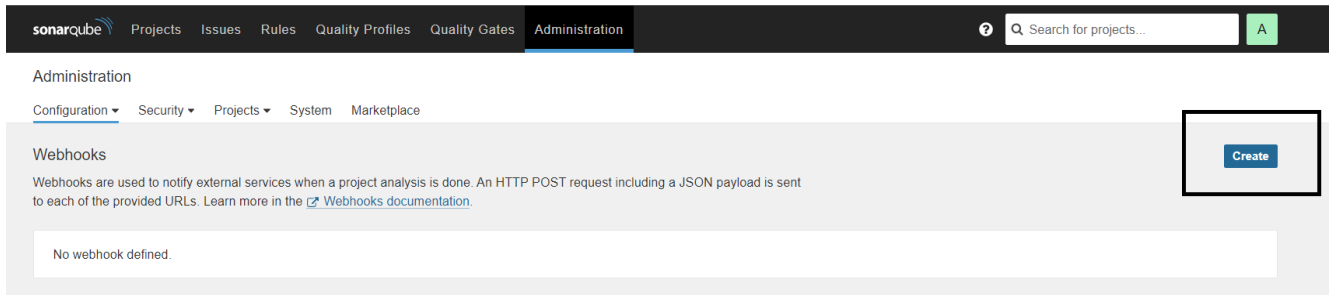
In the Sonarqube Dashboard add a quality gate also

Administration-> Configuration->Webhooks

The screenshot shows the SonarQube Administration interface. The top navigation bar includes 'Administration' which is highlighted with a red box. Below it, the 'Configuration' dropdown menu is open, and 'Webhooks' is selected and highlighted with a red box. The main content area shows a table of users. The first user is 'Administrator admin' with a green status icon. The table has columns for 'SCM Accounts', 'Last connection', 'Groups', and 'Tokens'. The 'Groups' column for the administrator user lists 'sonar-administrators' and 'sonar-users'. The 'Tokens' column shows '1' token and a settings icon.

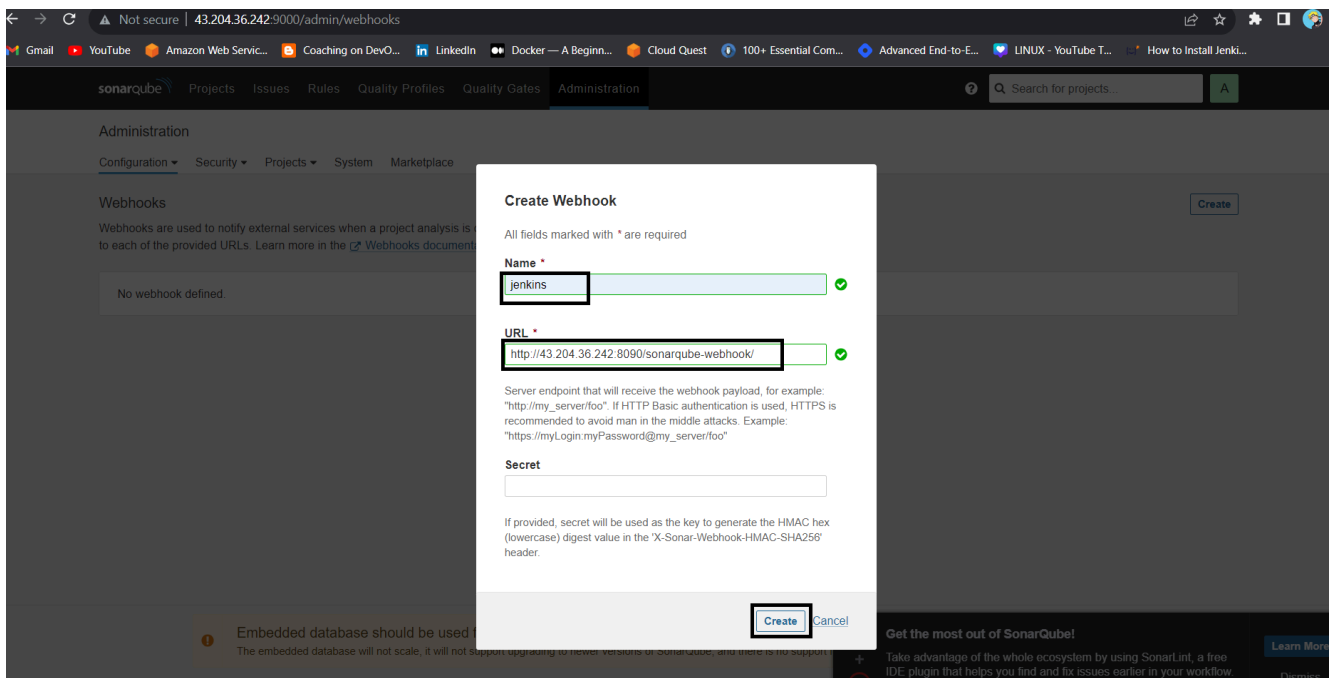
	SCM Accounts	Last connection	Groups	Tokens
Administrator admin		< 1 hour ago	sonar-administrators sonar-users	1

Click on Create



Add details

#in url section of quality gate
http://jenkins-public-ip:8080/sonarqube-webhook/



First, we configured the Plugin and next, we had to configure the Tool

Goto Dashboard → Manage Jenkins → Tools → Dependency-Check & Docker

Add Dependency-Check

**Dependency-Check**

Name

DP-Check



Install automatically ?

**Install from github.com**

Version

dependency-check 9.2.0

Add Installer ▾

Add Dependency-Check

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Docker installations

Add Docker

**Docker**

Name

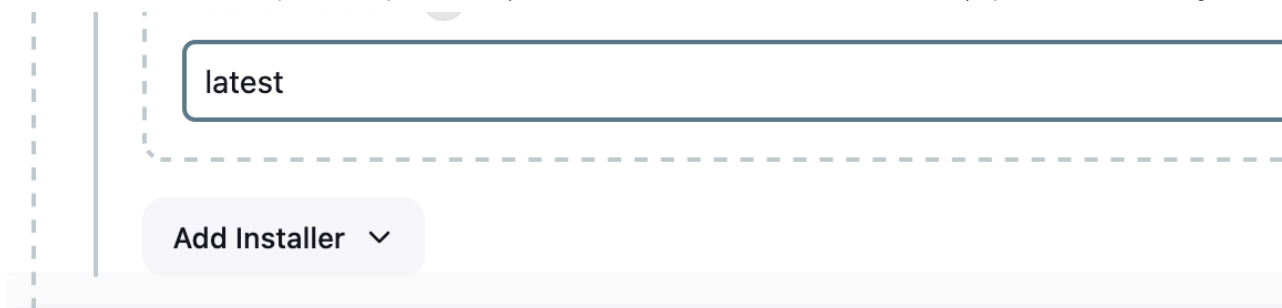
docker



Install automatically ?

**Download from docker.com**

Docker version ?



Go to manage Jenkins -> Credentials

Add DockerHub Username and Password under Global Credentials

A screenshot of the Jenkins 'Global credentials (unrestricted)' configuration page. The breadcrumb trail at the top reads: Dashboard > Manage Jenkins > Credentials > System > Global credentials (unrestricted) >. The form has the following fields: 'Kind' (dropdown menu with 'Username with password' selected), 'Scope' (dropdown menu with 'Global (Jenkins, nodes, items, all child items, etc)' selected), 'Username' (text input with 'sevenajay'), 'Treat username as secret' (checkbox, unchecked), 'Password' (password input with masked characters), 'ID' (text input with 'docker'), and 'Description' (text input with 'docker'). A blue 'Create' button is at the bottom left.

Create Job for Uptime-kuma

Add this stage to Pipeline Script



```
pipeline{
  agent any
  tools{
    jdk 'jdk17'
    nodejs 'node18'
  }
  environment {
    SCANNER_HOME=tool 'sonar-scanner'
  }
  stages {
```



```

stage('Checkout from Git'){
    steps{
        git branch: 'main', url: 'https://github.com/Aj7Ay/uptir
    }
}
stage('Install Dependencies') {
    steps {
        sh "npm install"
    }
}
stage("Sonarqube Analysis "){
    steps{
        withSonarQubeEnv('sonar-server') {
            sh ''' $SCANNER_HOME/bin/sonar-scanner -Dsonar.proje
            -Dsonar.projectKey=uptime '''
        }
    }
}
stage("quality gate"){
    steps {
        script {
            waitForQualityGate abortPipeline: false, credential:
        }
    }
}
stage('OWASP FS SCAN') {
    steps {
        dependencyCheck additionalArguments: '--scan ./ --disab
        dependencyCheckPublisher pattern: '**/dependency-check-r
    }
}
stage('TRIVY FS SCAN') {
    steps {
        sh "trivy fs . & trivyfs.json"
    }
}
stage("Docker Build & Push"){
    steps{
        script{
            withDockerRegistry(credentialsId: 'docker', toolName
            sh "docker build -t uptime ."
            sh "docker tag uptime sevenajay/uptime:latest "
            sh "docker push sevenajay/uptime:latest "
        }
    }
}

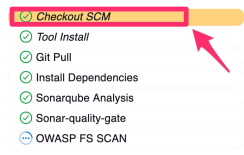
```

```

    }
}
stage("TRIVY"){
    steps{
        sh "trivy image sevenajay/uptime:latest &gt; trivy.json"
    }
}
stage ("Remove container") {
    steps{
        sh "docker stop uptime | true"
        sh "docker rm uptime | true"
    }
}
stage('Deploy to container'){
    steps{
        sh 'docker run -d --name chatbot -v /var/run/docker.sock'
    }
}
}
}

```

Git checkout



Stage 'Checkout SCM'

- Started 25 min ago
- Queued 0 ms
- Took 1.1 sec
- Success
- [View as plain text](#)

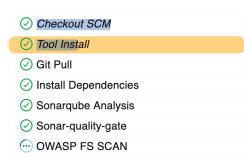
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```

0 Selected Git installation does not exist. Using Default
1 The recommended git tool is: NONE
2 No credentials specified
3 Cloning the remote Git repository
4 Cloning repository https://github.com/Aj7Ay/Uptime-kuma.git
5 > git init /var/lib/jenkins/workspace/Uptime # timeout=10
6 Fetching upstream changes from https://github.com/Aj7Ay/Uptime-kuma.git
7 > git --version # timeout=10
8 > git --version # 'git version 2.43.0'
9 > git fetch --tags --progress -- https://github.com/Aj7Ay/Uptime-kuma.git +refs/heads/*:refs/remotes/origin/* # timeout=10
10 > git config remote.origin.url https://github.com/Aj7Ay/Uptime-kuma.git # timeout=10
11 > git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
12 Avoid second fetch
13 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
14 Checking out Revision b010fe02f54dd9c7f3392be58dfc96946099e392 (refs/remotes/origin/main)
15 > git config core.sparsecheckout # timeout=10
16 > git checkout -f b010fe02f54dd9c7f3392be58dfc96946099e392 # timeout=10
17 Commit message: "Merge pull request #1 from Aj7Ay/feat/uptime-jenkins"
18 First time build. Skipping changelog.

```

Tool install

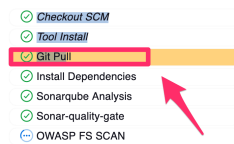


Stage 'Tool Install'

- Started 25 min ago
- Queued 0 ms
- Took 35 sec
- Success
- [View as plain text](#)

Tool	Action	Duration	Icons
jdk17	Use a tool from a predefined Tool Installation	23 sec	
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.		69 ms	
node18	Use a tool from a predefined Tool Installation	11 sec	
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.		51 ms	

Git pull



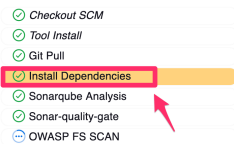
Stage 'Git Pull'

Started 25 min ago
Queued 0 ms
Took 1.2 sec
Success
[View as plain text](#)

jdk17	Use a tool from a predefined Tool Installation	40 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	38 ms	🔗	🔗	▼
node18	Use a tool from a predefined Tool Installation	59 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	89 ms	🔗	🔗	▼
Git		0.74 sec	🔗	🔗	▲

```
0 Selected Git installation does not exist. Using Default
1 The recommended git tool is: NONE
2 No credentials specified
3 > git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Uptime/.git # timeout=10
4 Fetching changes from the remote Git repository
5 > git config remote.origin.url https://github.com/Aj7Ay/Uptime-kuma.git # timeout=10
6 Fetching upstream changes from https://github.com/Aj7Ay/Uptime-kuma.git
7 > git --version # timeout=10
8 > git --version # 'git version 2.43.0'
9 > git fetch --tags --force --progress -- https://github.com/Aj7Ay/Uptime-kuma.git +refs/heads/*:refs/remotes/origin/* # timeout=10
10 > git rev-parse refs/remotes/origin/main^{commit} # timeout=10
11 Checking out Revision b010fe02f54dd9c7f3392be58dfc96946099e392 (refs/remotes/origin/main)
12 > git config core.sparsecheckout # timeout=10
13 > git checkout -f b010fe02f54dd9c7f3392be58dfc96946099e392 # timeout=10
14 > git branch -a -v --no-abbrev # timeout=10
15 > git checkout -b main b010fe02f54dd9c7f3392be58dfc96946099e392 # timeout=10
16 Commit message: "Merge pull request #1 from Aj7Ay/feat/uptime-jenkins"
```

NPM install

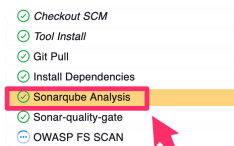


Stage 'Install Dependencies'

Started 25 min ago
Queued 0 ms
Took 1 min 49 sec
Success
[View as plain text](#)

jdk17	Use a tool from a predefined Tool Installation	45 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	47 ms	🔗	🔗	▼
node18	Use a tool from a predefined Tool Installation	54 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	82 ms	🔗	🔗	▼
npm install	Shell Script	1 min 48 sec	🔗	🔗	▼

Sonarqube analysis



Stage 'Sonarqube Analysis'

Started 23 min ago
Queued 0 ms
Took 2 min 31 sec
Success
[View as plain text](#)

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jdk17	Use a tool from a predefined Tool Installation	43 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	32 ms	🔗	🔗	▼
node18	Use a tool from a predefined Tool Installation	44 ms	🔗	🔗	▼
	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	58 ms	🔗	🔗	▼
SSCANNER_HOME/bin/sonar-scanner -Dsonar.projectName=Chatbot -Dsonar.projectKey=Chatbot	Shell Script	2 min 28 sec	🔗	🔗	▼

☆ Chatbot

Passed

Last analysis: 2 hours ago

🐛 Bugs	🔒 Vulnerabilities	🔥 Hotspots Reviewed	👤 Code Smells	Coverage	Duplications	Lines
41 E	0 A	0.0% E	277 A	0.0% E	1.5% A	31k M JavaScript...

Quality Gate



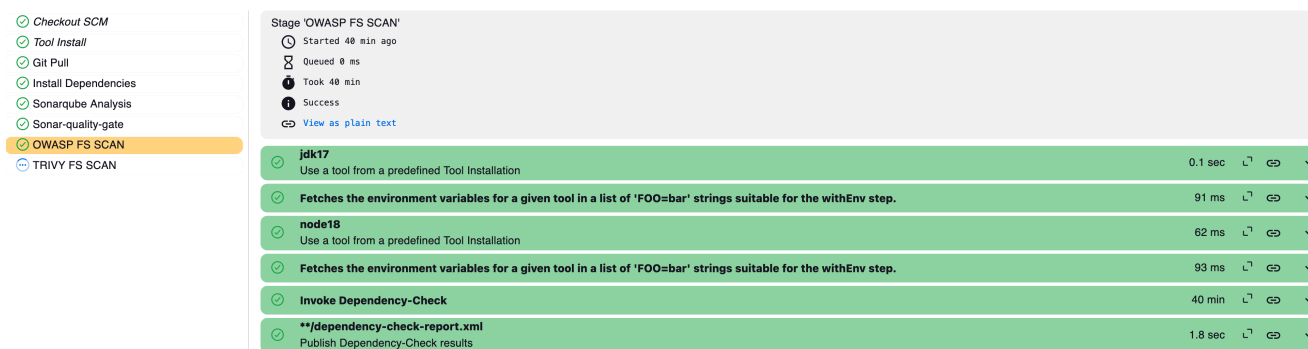
Stage 'Sonar-quality-gate'

- Started 21 min ago
- Queued 0 ms
- Took 7.6 sec
- Success
- [View as plain text](#)

Step	Description	Duration	Actions
✓	jdk17 Use a tool from a predefined Tool Installation	55 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	62 ms	L C D V
✓	node18 Use a tool from a predefined Tool Installation	0.11 sec	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	81 ms	L C D V
✓	Wait for SonarQube analysis to be completed and return quality gate status	6.9 sec	L C D V ^

```
0 Checking status of SonarQube task 'AZBSrEmIUMbsm6jdmkwe' on server 'sonar-server'
1 SonarQube task 'AZBSrEmIUMbsm6jdmkwe' status is 'IN_PROGRESS'
2 SonarQube task 'AZBSrEmIUMbsm6jdmkwe' status is 'SUCCESS'
3 SonarQube task 'AZBSrEmIUMbsm6jdmkwe' completed. Quality gate is 'OK'
```

Owasp

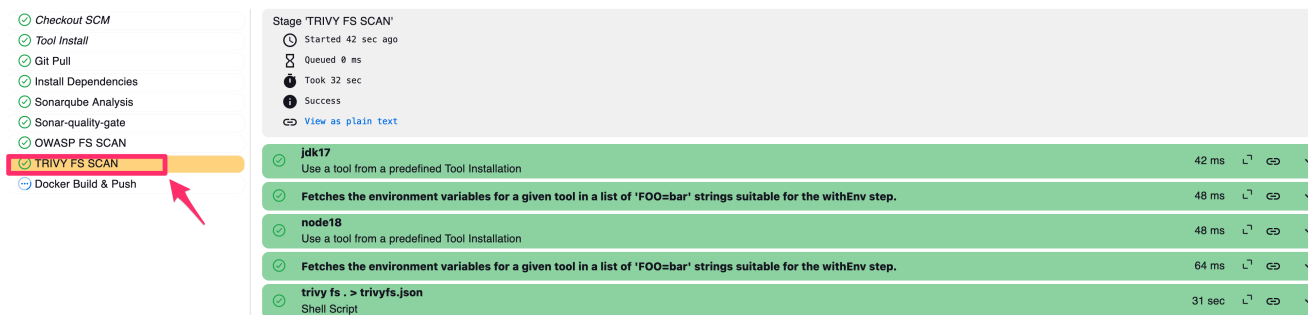


Stage 'OWASP FS SCAN'

- Started 40 min ago
- Queued 0 ms
- Took 40 min
- Success
- [View as plain text](#)

Step	Description	Duration	Actions
✓	jdk17 Use a tool from a predefined Tool Installation	0.1 sec	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	91 ms	L C D V
✓	node18 Use a tool from a predefined Tool Installation	62 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	93 ms	L C D V
✓	Invoke Dependency-Check	40 min	L C D V
✓	**/dependency-check-report.xml Publish Dependency-Check results	1.8 sec	L C D V

Trivy FS

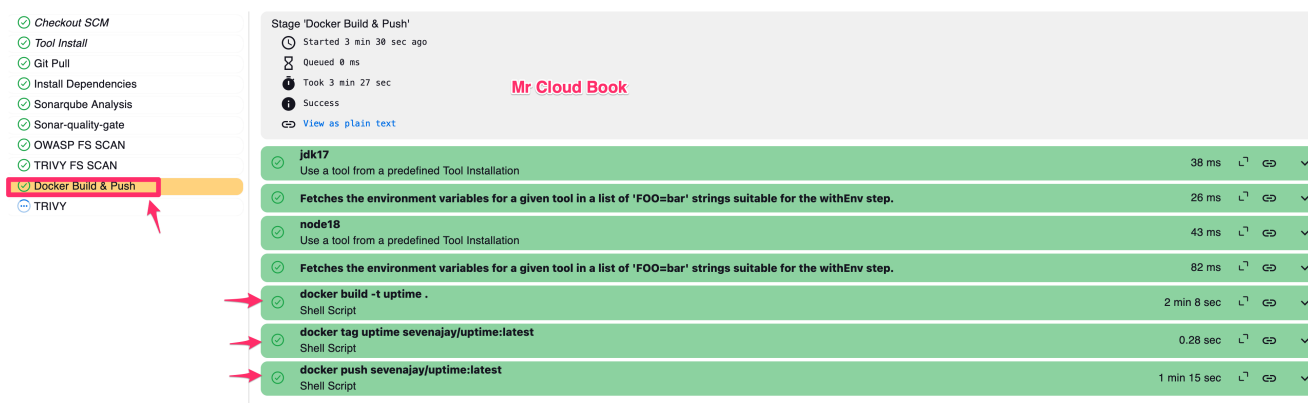


Stage 'TRIVY FS SCAN'

- Started 42 sec ago
- Queued 0 ms
- Took 32 sec
- Success
- [View as plain text](#)

Step	Description	Duration	Actions
✓	jdk17 Use a tool from a predefined Tool Installation	42 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	48 ms	L C D V
✓	node18 Use a tool from a predefined Tool Installation	48 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	64 ms	L C D V
✓	trivy fs . > trivyfs.json Shell Script	31 sec	L C D V

Docker Build and push



Stage 'Docker Build & Push'

- Started 3 min 30 sec ago
- Queued 0 ms
- Took 3 min 27 sec
- Success
- [View as plain text](#)

Step	Description	Duration	Actions
✓	jdk17 Use a tool from a predefined Tool Installation	38 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	26 ms	L C D V
✓	node18 Use a tool from a predefined Tool Installation	43 ms	L C D V
✓	Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	82 ms	L C D V
✓	docker build -t uptime . Shell Script	2 min 8 sec	L C D V
✓	docker tag uptime sevenajay/uptime:latest Shell Script	0.28 sec	L C D V
✓	docker push sevenajay/uptime:latest Shell Script	1 min 15 sec	L C D V

Trivy Image scan and deploy to Docker

The screenshot shows a CI/CD pipeline interface. On the left, a list of steps is displayed, with 'TRIVY' highlighted in orange and a red box around it. On the right, the 'TRIVY' stage is expanded, showing a list of steps with their durations and status.

Step	Duration	Status
jdk17	41 ms	Success
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	34 ms	Success
node18	63 ms	Success
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.	57 ms	Success
trivy image sevenajay/uptime:latest > trivy.json	4 min 14 sec	Success

Copy the Public of Server



public-ip>;:30001



Output Login Page



Uptime Kuma

Create your admin account

Language
English



Username
mrcloudbook



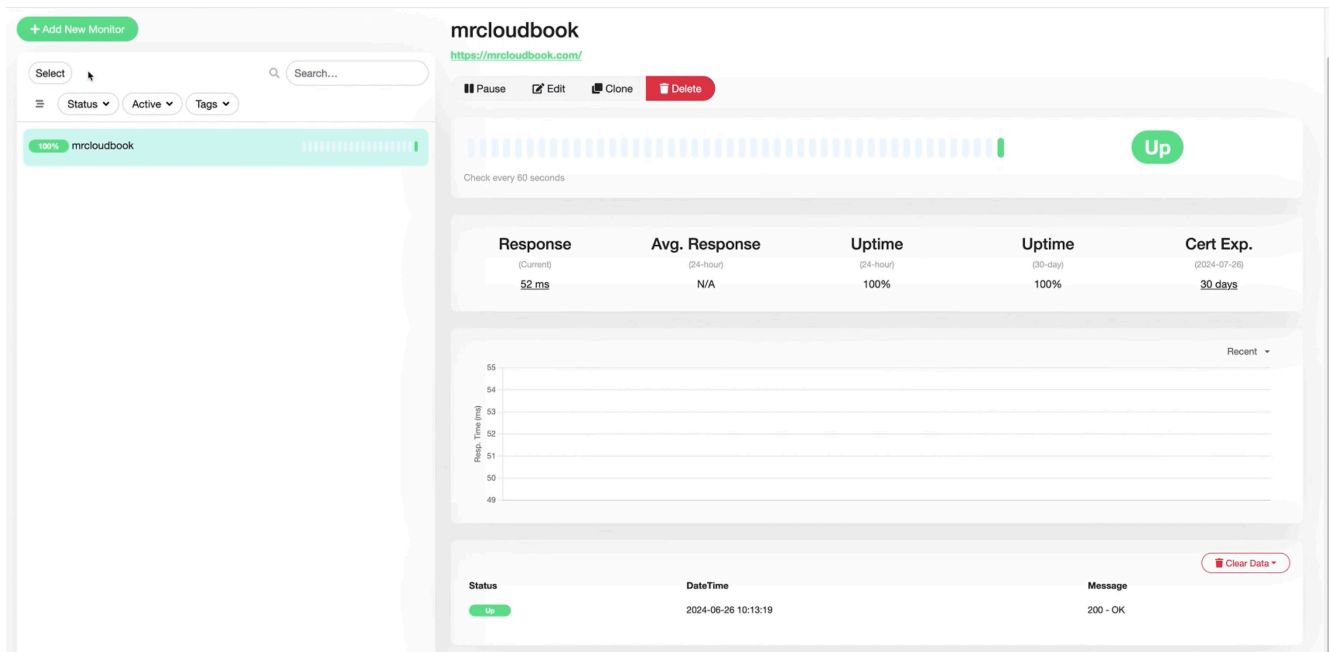
Password

Repeat Password

Create



Output



Twilio webhook setup

Clone the Github Repo

```
git clone https://github.com/Aj7Ay/Uptime-kuma.git
cd Uptime-kuma
cd Webhook
```

Create an Twilio Account From [Here](#)

Go to Console

copy Account SID & Account Auth Token and Mobile Number

Console
My first Twilio account Trial: \$5.4518 Upgrade

Develop Monitor

> # Phone Numbers
> Messaging
> Studio
> Voice
> Conversations

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Ahoy Ajay, welcome to Twilio!

Connect to 3rd-party applications

You'll need 3 things to use Twilio with most 3rd-party applications:

- ✓ Account SID and Auth token
- ✓ Twilio phone number
- ✓ Upgraded Twilio account

[Upgrade your account](#) →

[Read 3rd-party integration FAQ](#)

Account Info

Account SID
AC5a3f... [Show](#)

Auth Token
..... [Show](#)

⚠ Always store your token securely to protect your account. [Learn more](#)

My Twilio phone number
+1985... [Show](#)

You are in a trial account and can only send messages and make calls to [verified phone numbers](#). [Learn more about your trial account](#)

Update the `app.py`.



```
from flask import Flask
from twilio.rest import Client

app = Flask(__name__)

@app.route('/trigger-calls', methods=['GET', 'POST'])
def trigger_call():
    account_sid = '<acc-sid>'
    auth_token = '<acc-auth>'
    client = Client(account_sid, auth_token)

    # List of numbers to call
```



```
numbers_to_call = ['&country-code-number&','&country-code-  
call_sid = [] # To store Call SIDs  
  
for number in numbers_to_call:  
    call = client.calls.create(  
        twim='&Response&&Say&',"change Al<  
        to=number,  
        from_='&twilio-number&'  
    )  
    call_sids.append(call.sid)  
  
# Joining all the Call SIDs to return in response  
call_sids_str = ', 'join(call_sids)  
return f"Call initiated with SIDs: {call_sids_str}", 200  
  
if __name__ == '__main__':  
    app.run(host'0.0.0.0', port=5000)
```

Now install These on Server



```
sudo apt install python3-pip  
sudo apt install python3.12-venv  
python3 -m venv venv  
source venv/bin/activate
```



Now install Requirements



```
pip install -r requirements.txt
```



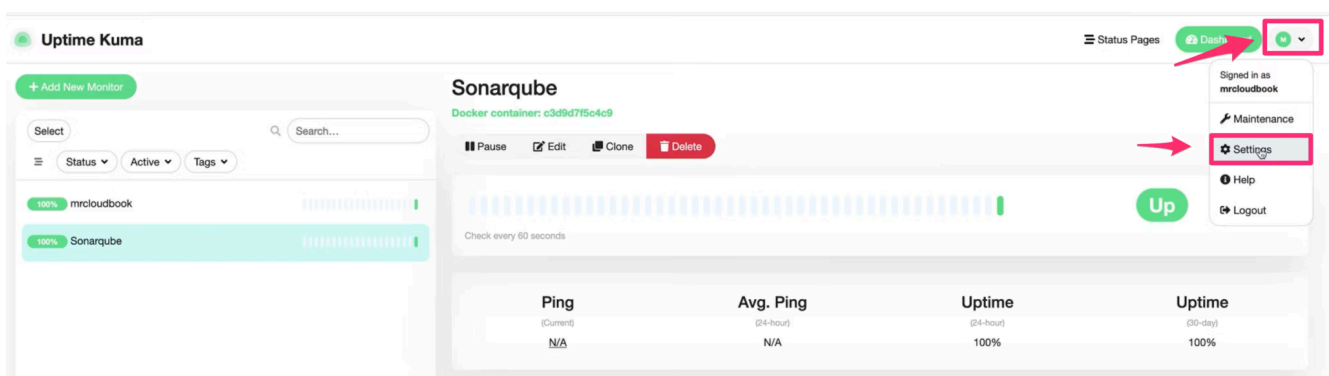
Create the Docker webhook



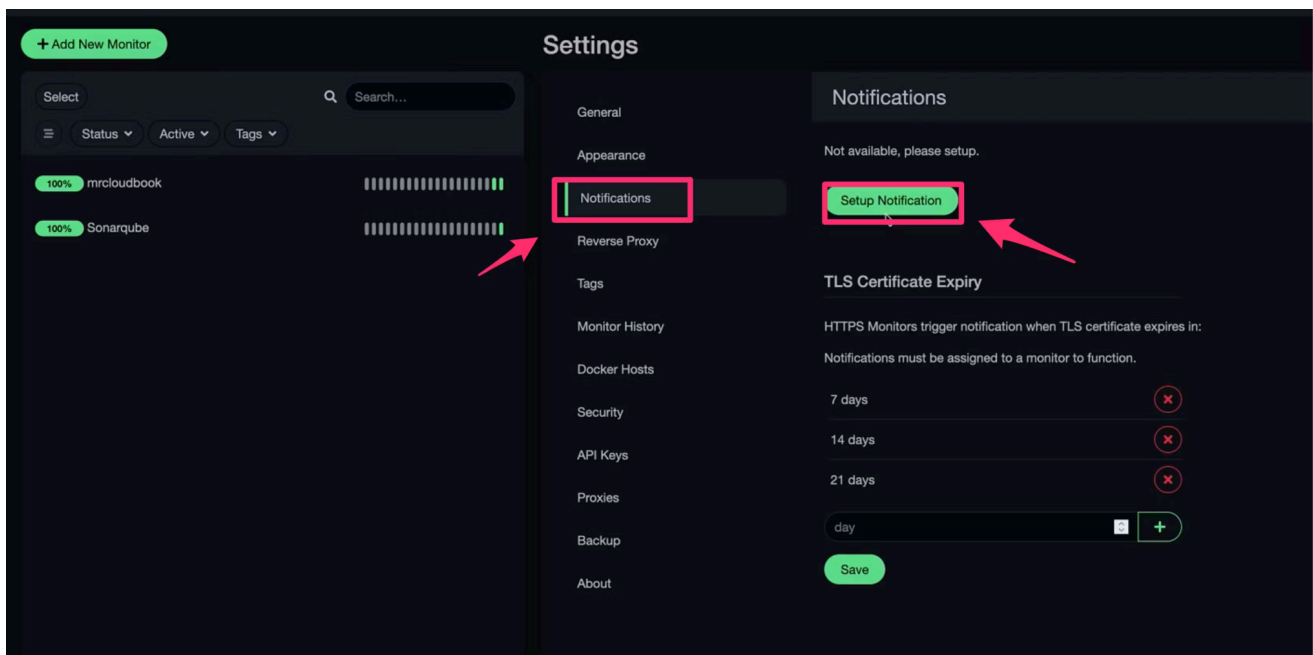
```
cd Uptime-kuma/webhook
docker build -t webhook .
docker run -d --name webhook -p 5000:5000 webhook:latest
```

webhook runs on 5000 port

Go to Uptime Kuma select Profile -> Settings



Select Notification -> setup Notification



Select Notification Type as webhook

Give Friendly name -> URL of webhook -> save

Setup Notification ✕

Notification Type

Webhook

Friendly Name

Twilio

Post URL

http://65.0.93.191:5000//trigger-call

Request Body

Preset - application/json

application/json is good for any modern HTTP servers such as Express.js

☐ Additional Headers

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Sets additional headers sent with the webhook. Each header should be defined as a JSON key/value.

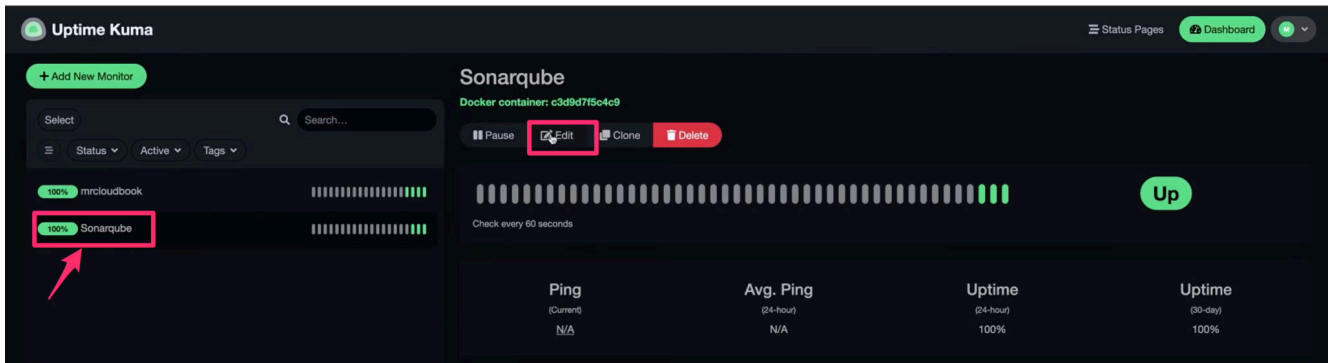
☐ Default enabled

This notification will be enabled by default for new monitors. You can still disable the notification separately for each monitor.

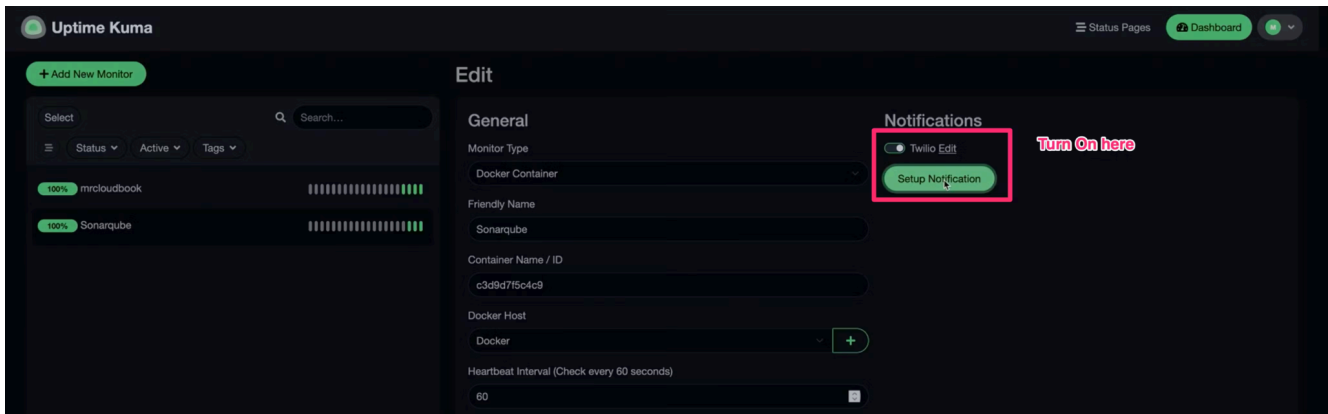
☐ Apply on all existing monitors

Test Save

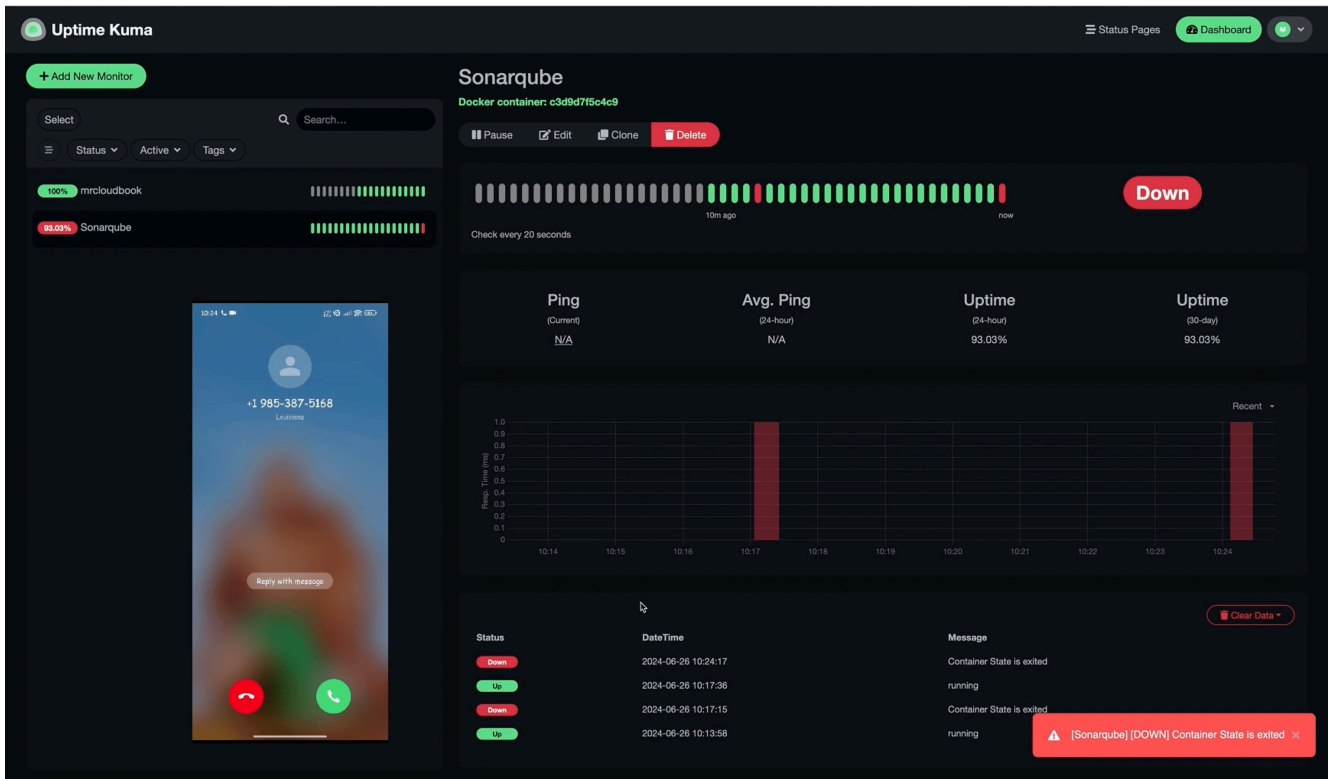
Select One service from Dashboard -> Edit



Turn on Notification to get call



You will get call when server or service Down



Remove Server at end.



Ajay Kumar Yegireddi is a DevSecOps Engineer and System Administrator, with a passion for sharing real-world DevSecOps projects and tasks. **Mr. Cloud Book**, provides hands-on tutorials and practical insights to help others master DevSecOps tools and workflows. Content is designed to bridge the gap between development, security, and operations, making complex concepts easy to understand for both beginners and professionals.

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18 July 2024

Hellpo

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